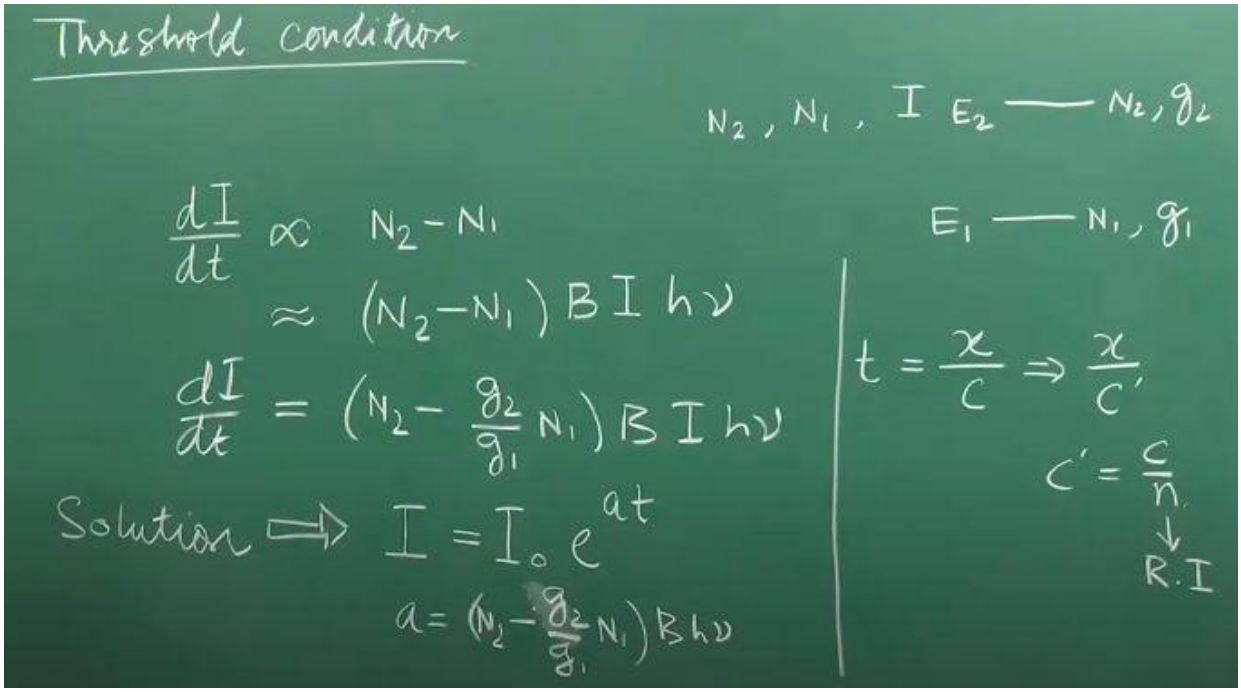
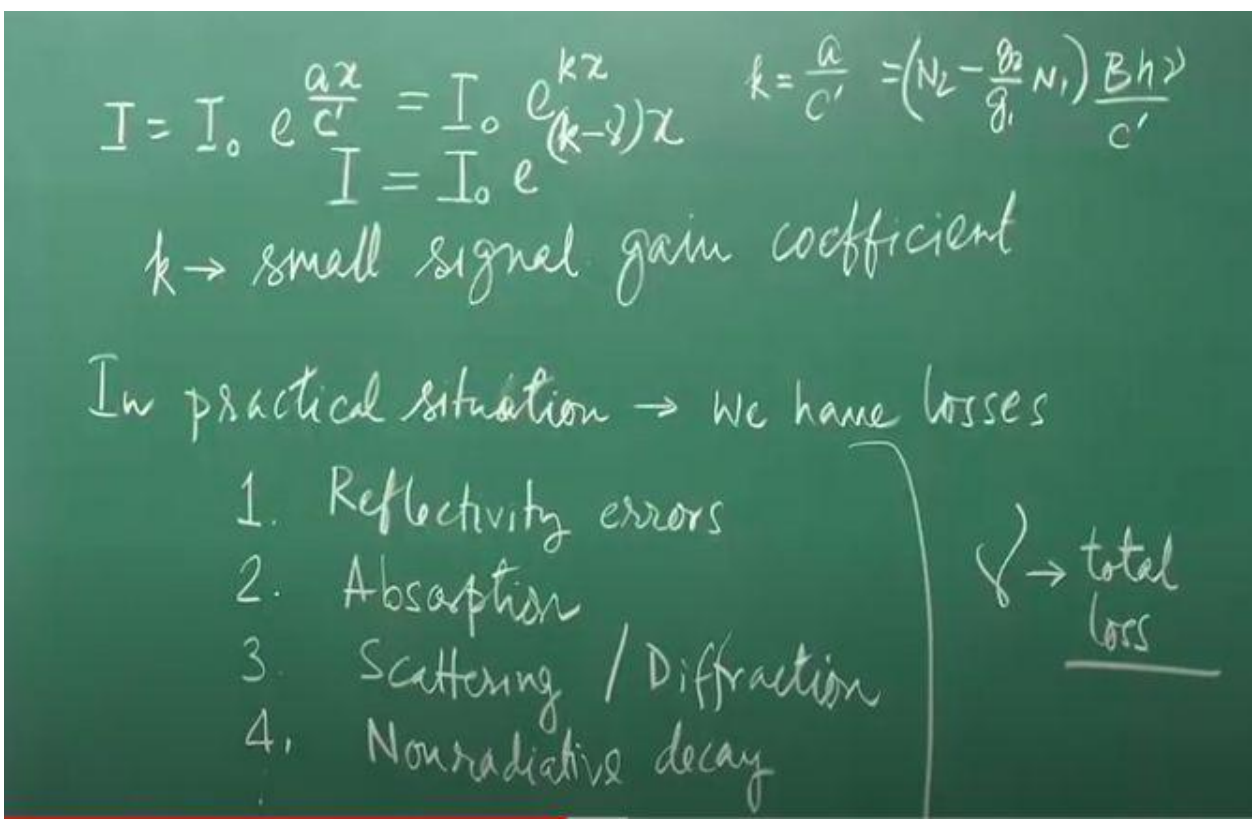
I = Intensity of stimulated emission

How did he write $(N_2 - N_1) = (N_2 - g_2/g_2 * N_1)$ where g_1, g_2 are degeneracy of a system. I.e., no, of states in a given energy level.





Threshold condition

$R_1 \neq R_2$

100% 98%

Gain $I = I_0 e^{(k-\gamma)x} \times R_1 \times R_2$

$$G = \frac{I}{I_0} = R_1 R_2 e^{(k-\gamma)x}$$

For laser action (true amplification)

$$G > 1$$

If $G < 1 \longrightarrow$ Work like a conventional light source

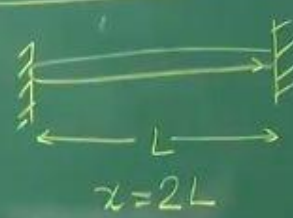
How to find the Threshold cond.? $\Rightarrow G = 1$

21:16 / 28:10

$$I = I_0 e^{\frac{a x}{c'}} = I_0 e^{kx}$$

$$I = I_0 e^{(k-\gamma)x}$$

$$\boxed{k = \frac{a}{c'} = \left(N_2 - \frac{g_2}{g_1} N_1\right) \frac{B h \nu}{c'}}$$



$$R_1 R_2 e^{(k-\gamma)x} = 1$$

$$(k-\gamma)x = \log_e\left(\frac{1}{R_1 R_2}\right)$$

$$(k-\gamma)2L = \log_e\left(1/R_1R_2\right)$$

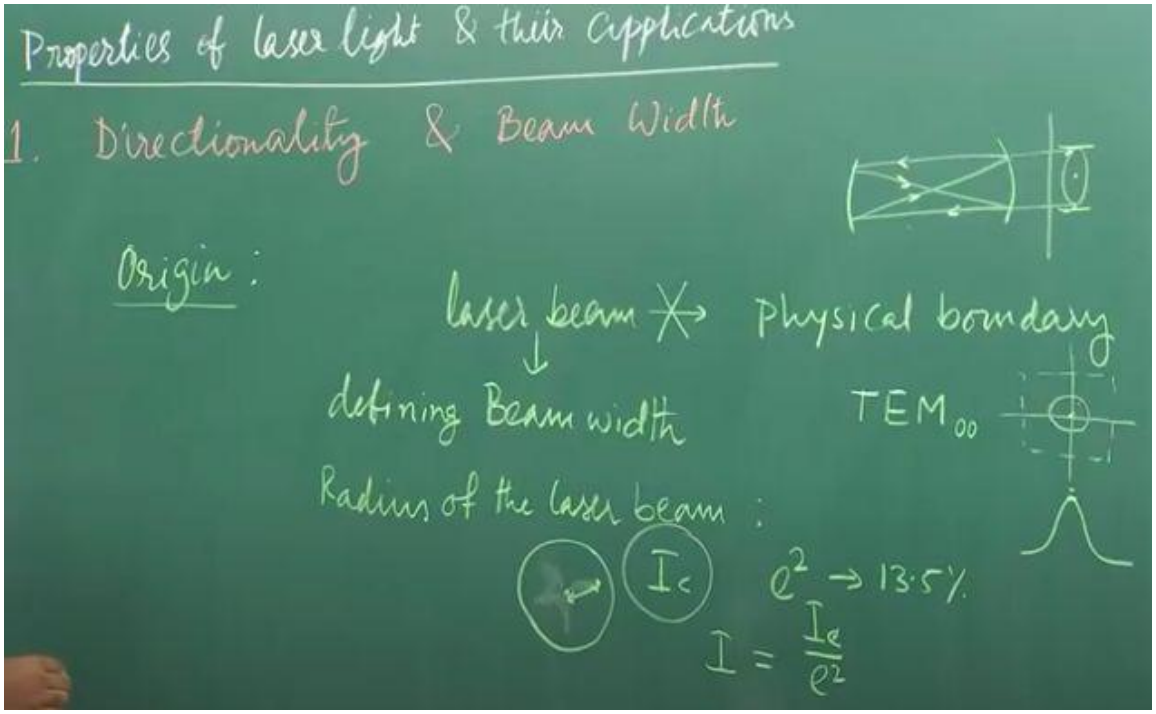
$$\boxed{k_{threshold} = \gamma + \frac{1}{2L} \log_e\left(1/R_1R_2\right)}$$

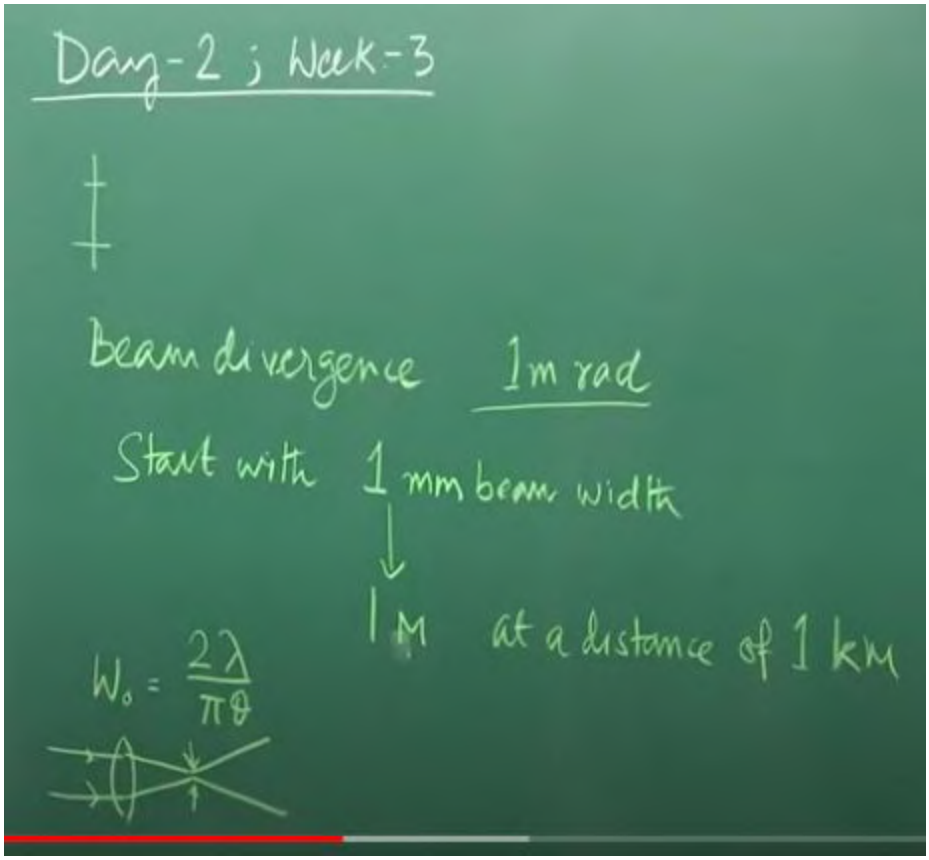
$$\left(N_2 - \frac{g_2}{g_1} N_1\right) \frac{B h \nu}{c'} > \left[\gamma + \frac{1}{2L} \log_e\left(\frac{1}{R_1 R_2}\right)\right]$$

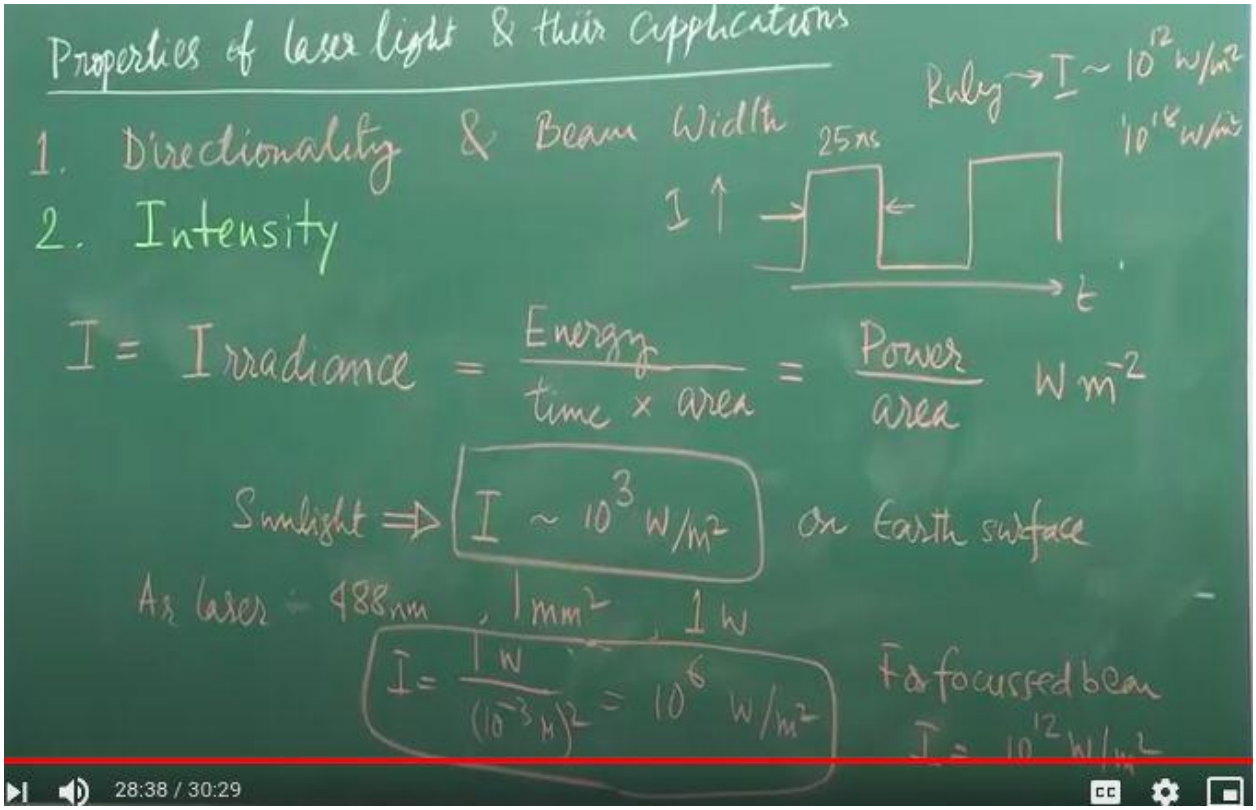
True : LASER output

5:36 / 28:10

Directionaly is used for alignment in Industries.
Intensity is used to cut, weld and mel in Industries

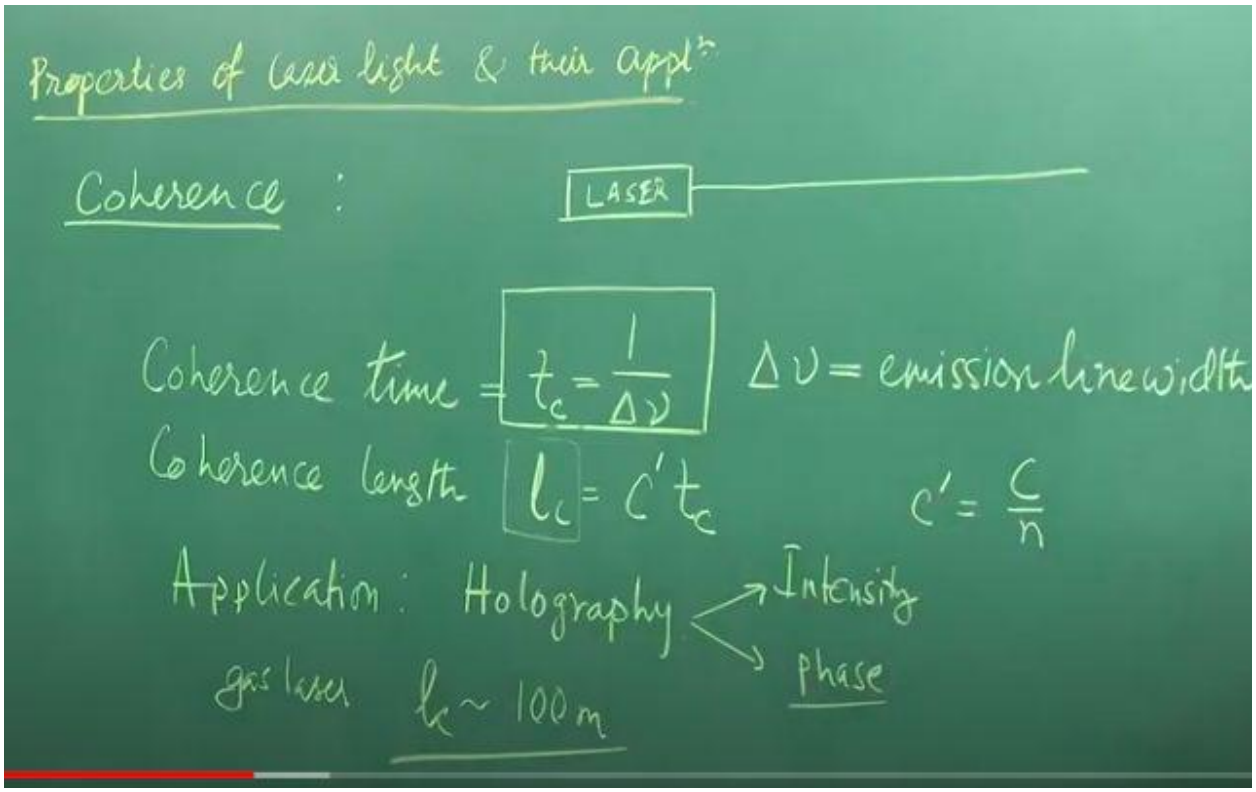


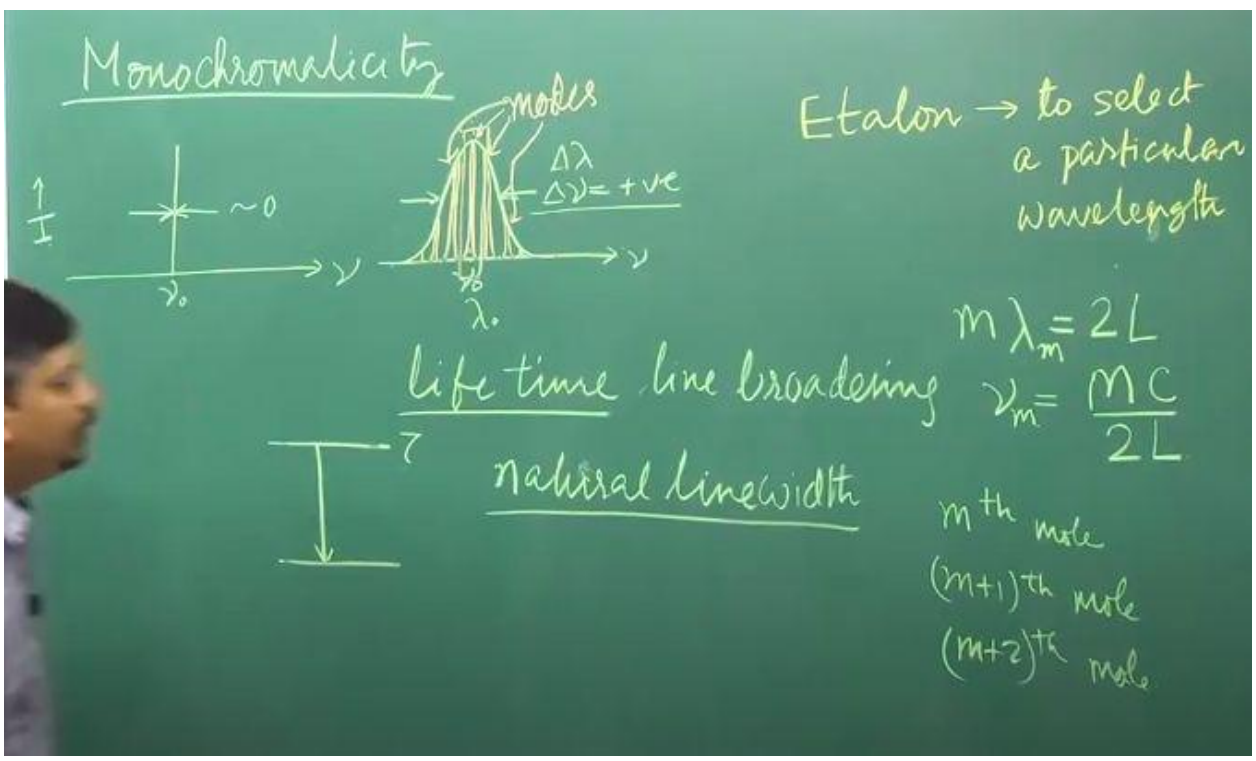



Properties of laser light & their applications
1. Directionality & Beam Width
2. Intensity
Ruby → I ~ 10^12 W/m^2
10^18 W/m^2
25ns
I
t
I = Irradiance = Energy / time × area = Power / area W m^-2
Sunlight ⇒ I ~ 10^3 W/m^2 on Earth surface
Ar laser 488nm , 1 mm^2 , 1 W
I = 1 W / (10^-3 m)^2 = 10^6 W/m^2
For focussed beam
I = 10^12 W/m^2
28:38 / 30:29

.

Narrow linewidth laser are used in nuclear industry to separate isotopes
Coherence for holograms and OCT





Monochromaticity

$L = 100\ cm$

$\lambda = 500\ nm = 500 \times 10^{-7}\ cm$

$$m = \frac{2L}{\lambda} = \frac{2 \times 100\ cm}{500 \times 10^{-7}\ cm} = 4 \times 10^{6} = 4000000^{th}$$

Gap between two successive modes

$$\Delta\nu = \frac{c}{2L} \Leftarrow \boxed{\nu_m = \frac{mc}{2L}} = 6 \times 10^{14}\ Hz$$

$4000001^{th} \Rightarrow \lambda\ ?$ $\nu\ ?$

$$\lambda_{(m+1)} = \frac{2L}{(m+1)} = \frac{2 \times 100}{4000001} = 499.999\ nm$$

Properties of laser light & their appl^n

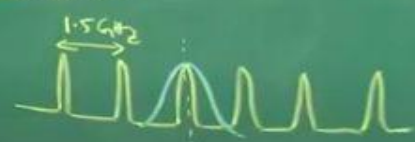
He Ne : bandwidth $1.5\ GHz$

$L = 10\ cm$

$$\Delta\nu_{HeNe} = \frac{c\ cm/s}{2 \times 10\ cm} = 1.5 \times 10^{9}\ Hz = 1.5\ GHz$$

$L = 30\ cm$

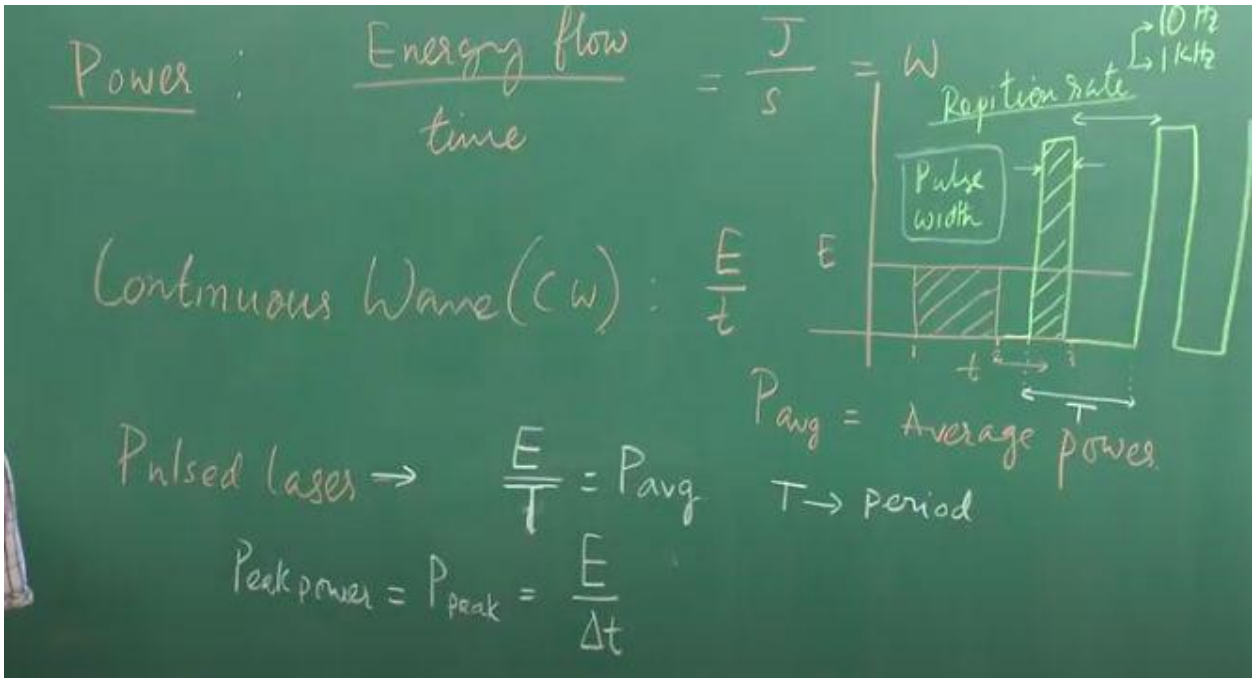
$\Delta\nu = 0.5\ GHz$

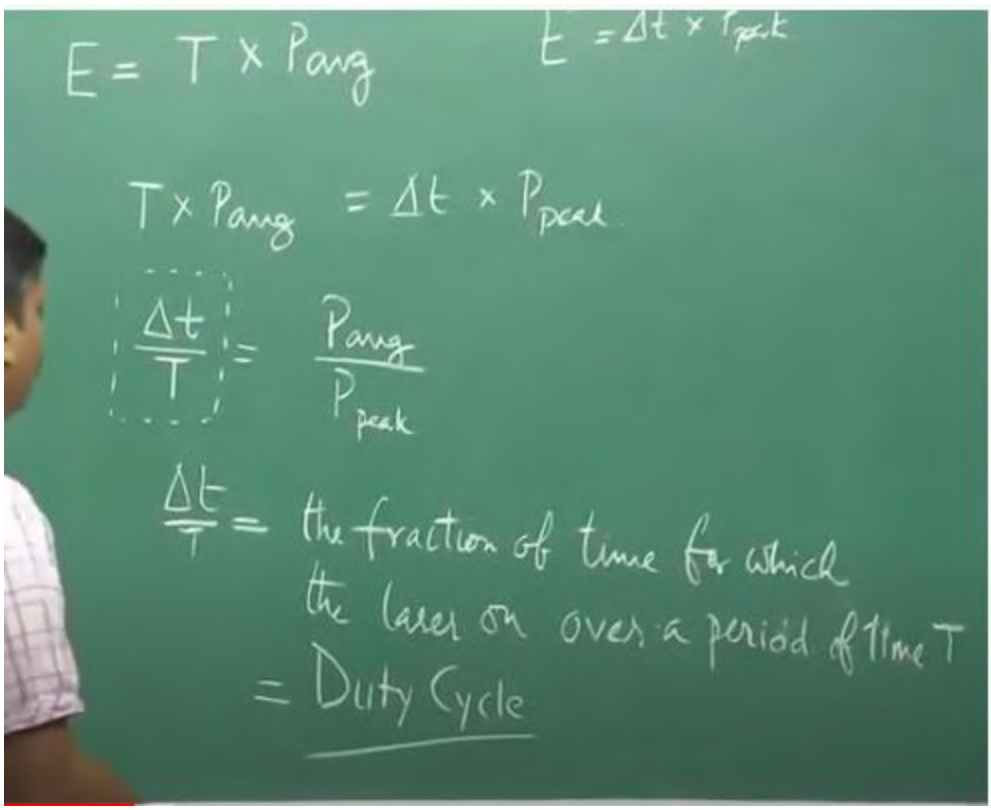


$$\frac{1.5\ GHz}{0.5\ GHz} = 3\ modes$$

Quality factor

$$Q = \frac{\nu}{\Delta\nu} \Rightarrow 10^{8}$$


Power : Energy flow / time = J/s = W
10 Hz
1 kHz
Repition rate
Pulse width
E
t
T
Continuous Wave (CW) : E/t
P_avg = Average power
Pulsed lases → E/T = P_avg T → period
Peak power = P_peak = E/Δt

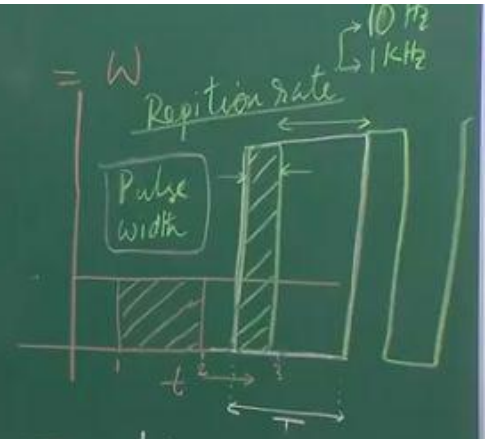

E = T × P_avg
E = Δt × P_peak
T × P_avg = Δt × P_peak
Δt/T = P_avg/P_peak
Δt/T = the fraction of time for which the laser on over a period of time T
= Duty Cycle

Example

A Laser: Peak power 10 kW

Pulse width $1\ ns = 10^{-9}\ s$

Repition rate 1 kHz



$T \sim 1/1000$

$$P_{avg} = \frac{\Delta t}{T} \times P_{peak}\ W$$

$$= \frac{10^{-9}}{1/1000} \times 10 \times 10^{3}\ W$$

$$= 10^{-6} \times 10^{4} = 10^{-2}\ W \rightarrow 10\ mW$$

Pulse width

Pulse-shape $\rightarrow$ Gaussian $\rightarrow P(t) = P_p\, e^{-4\ln 2 (t/\tau_p)^2}$

$\rightarrow$ $sech^2$

$\rightarrow$ Arbitrary

Pulse-shaping

Time-bandwidth product:

$$\tau_p \times \Delta\nu \approx 0.31$$

$sech^2$

Transform limit

$\downarrow$

Transform limited pulse